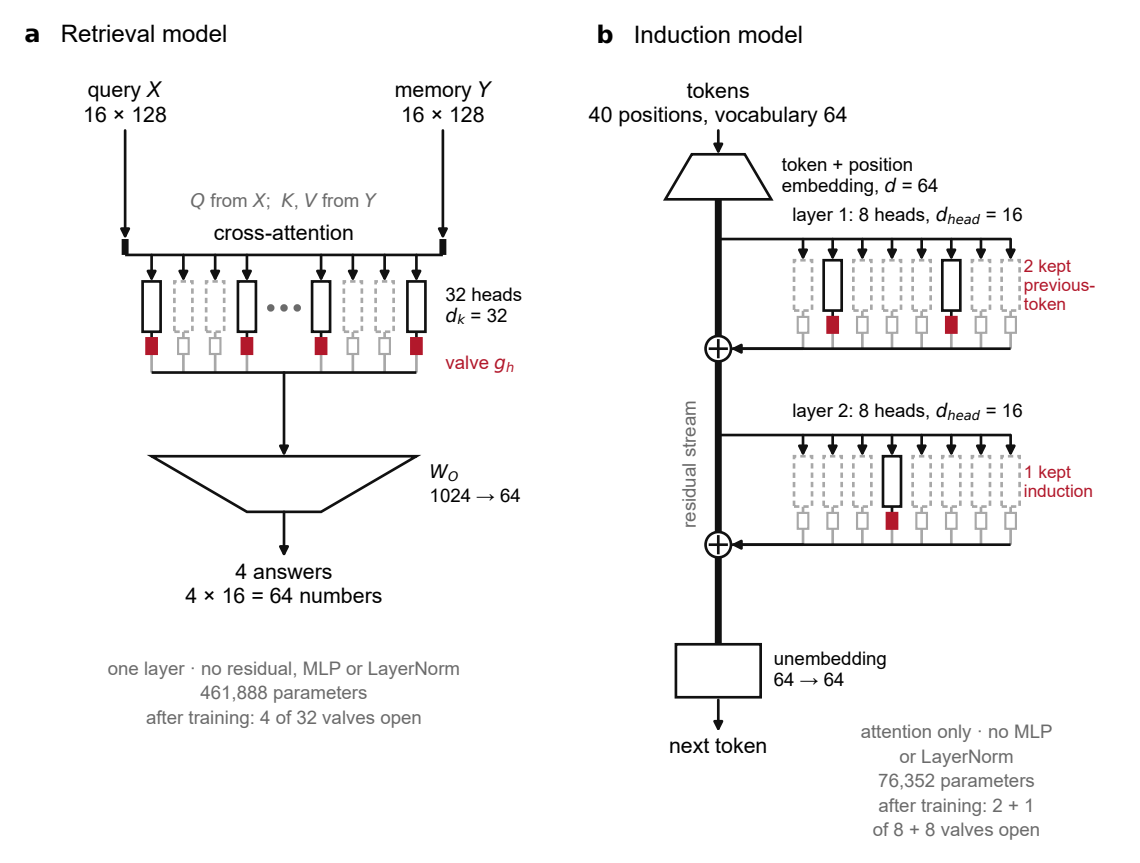
a Retrieval model
query X
16 × 128
memory Y
16 × 128
Q from X; K, V from Y
cross-attention
32 heads
d_k = 32
valve g_h
W_O
1024 → 64
4 answers
4 × 16 = 64 numbers
one layer · no residual, MLP or LayerNorm
461,888 parameters
after training: 4 of 32 valves open
b Induction model
tokens
40 positions, vocabulary 64
token + position
embedding, d = 64
layer 1: 8 heads, d_{head} = 16
2 kept
previous-
token
residual stream
layer 2: 8 heads, d_{head} = 16
1 kept
induction
unembedding
64 → 64
next token
attention only · no MLP
or LayerNorm
76,352 parameters
after training: 2 + 1
of 8 + 8 valves open